



MASSACHUSETTS
GENERAL HOSPITAL



HARVARD
MEDICAL SCHOOL

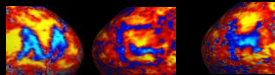
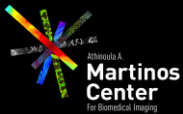


Massachusetts
Institute of
Technology

MR physics for fMR

HST 583 (physics lecture #1)

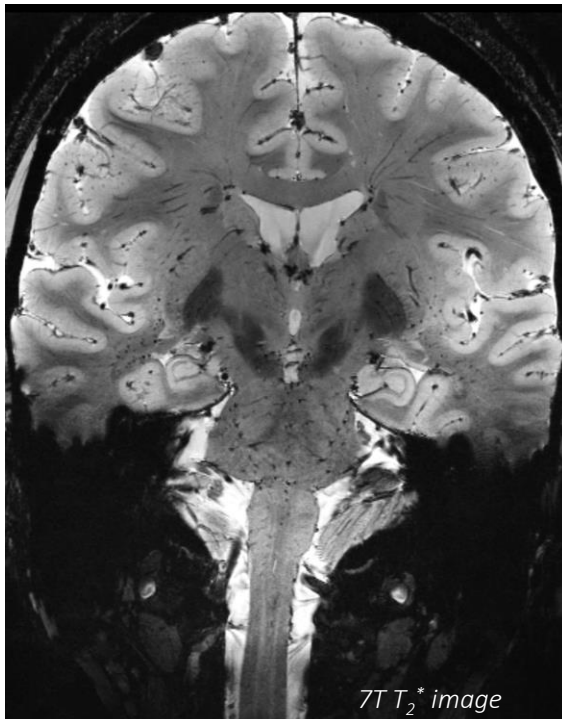
Lawrence L. Wald



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Disclosures:

- Equity, consulting, SAB: Neuro42 Inc.
- Equity, InnoCerebra, Reveal Pharma
- Research support:
Siemens Healthcare
- Will discuss non-FDA approved devices



7T T_2^* image



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Outline:

MR physics lecture 1:

MR signal,
Gradient and spin echo
T2*, T2, and T1 contrast
BOLD contrast basics

MR physics lecture 2:

Encoding the image

MR physics lecture 3:

Encoding review
Fast imaging fMRI artifacts
Alternatives to BOLD

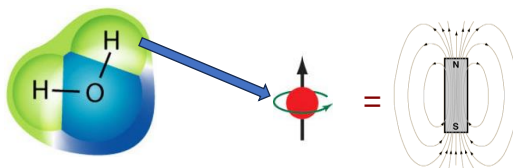


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What is NMR?

NUCLEAR MAGNETIC RESONANCE

A magnet, a glass of water,
and a radio wave source and detector....



Water magnetism:

Is water magnetic? Yes, weakly

Orbital electrons, electron spin:

Diamagnetism; no un-paired e⁻,
quenched orbital angular momentum.

Nuclear magnetism:

O¹⁶ (even number of p and n
> **no magnetic moment**)

H¹ (proton, spin ½) > magnetic moment

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History aside

Spins and nuclear magnetism

1922 Stern-Gerlach flip spins using RF, show quant. E, using atomic beams.

1926 Pauli, full spin theory in QM

1932 Stern shows nuclei have spin.

1936 Gorter predicts NMR, fails to detect (T_1 was too long...)

1938 I.I. Rabi, did NMR in beams, measured magnetic moments of nuclei

1946 Purcell, Pound, NMR in bulk material, via cw absorption of RF energy

1946 Bloch, "Nuclear Induction" (NMR in bulk, Bloch eq.), cw but viewed as Faraday induction

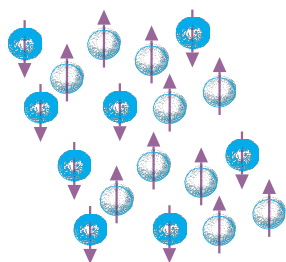
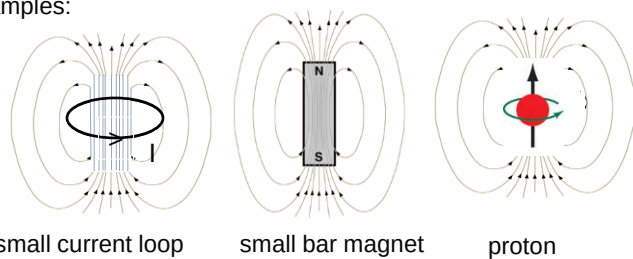
During WWII, Rabi and Purcell worked together on radar at MIT, while Bloch was at Harvard...

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Some house-keeping

Magnetic dipole is a particular magnetic field pattern.

examples:



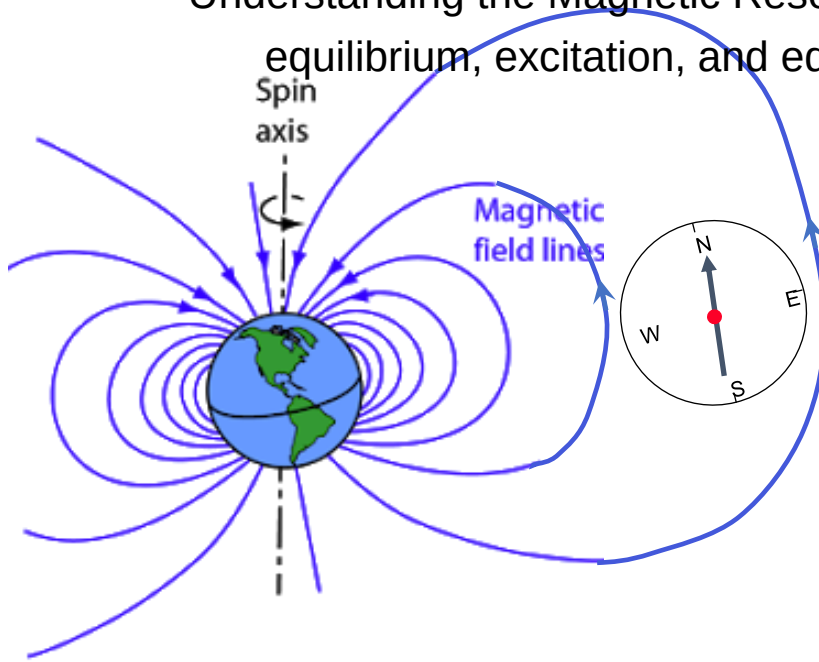
Net alignment:

$$\vec{M} = \sum \vec{m}$$

We manipulate & observe this macroscopic ensemble

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Understanding the Magnetic Resonance phenomena: equilibrium, excitation, and equation of motion

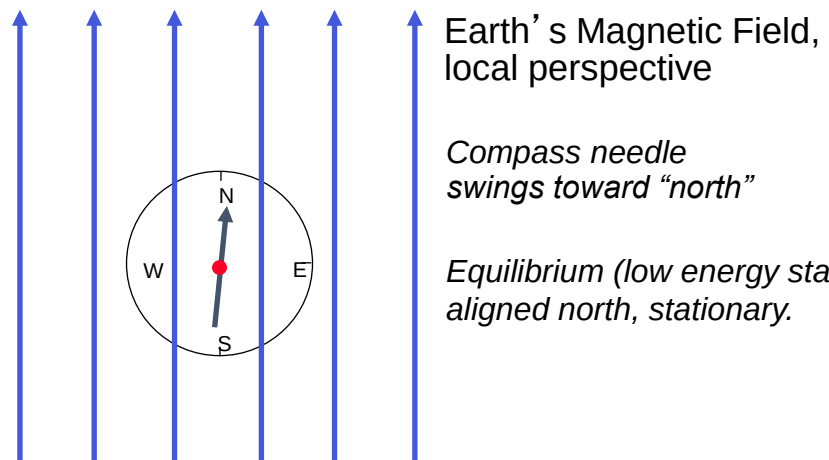


Compass,
A little magnet that
aligns toward "north"

How does it move
when you flick it?

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Compass needle (little magnet) in the earth's field



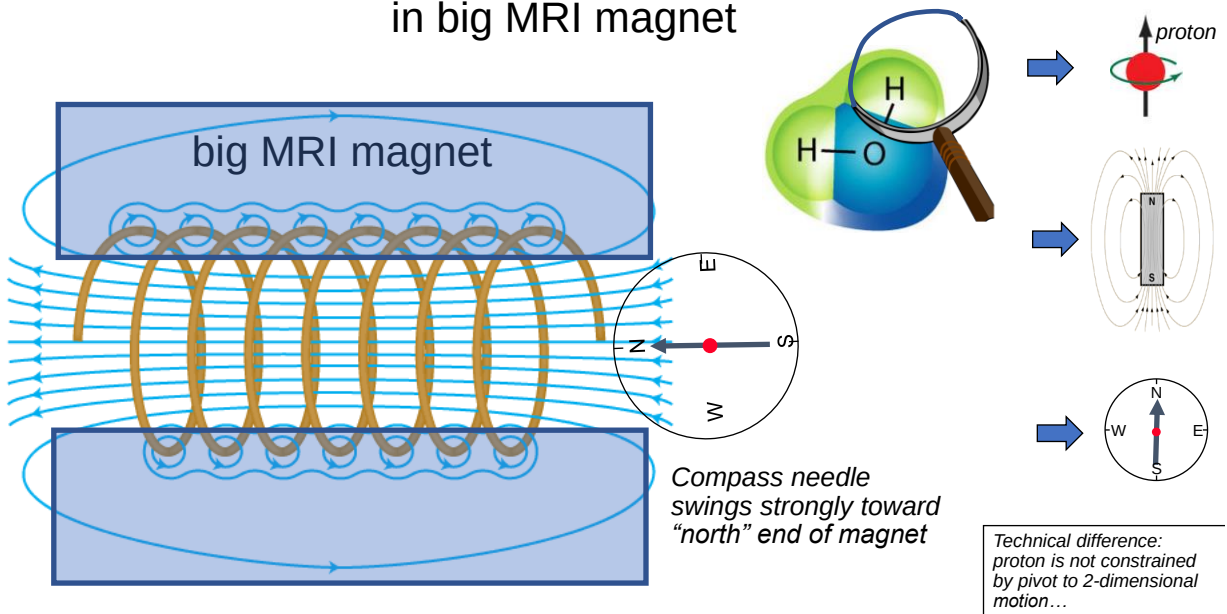
Earth's Magnetic Field,
local perspective

*Compass needle
swings toward "north"*

*Equilibrium (low energy state):
aligned north, stationary.*

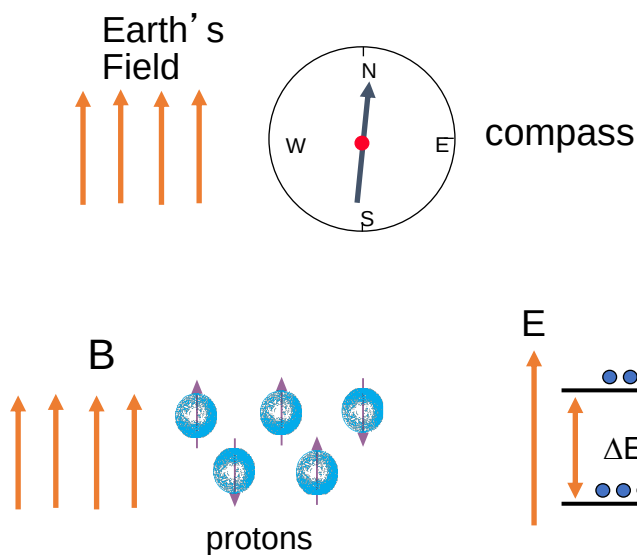
8

Protons (little magnets) in water do the same
in big MRI magnet



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Magnetic moment in an external B field



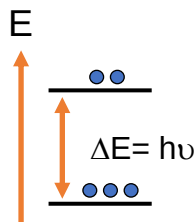
Two ways of saying the same thing:

- 1) The field torques the needle (magnetization) into alignment.
- 2) Alignment along north is the lowest energy state: $E = -\cos \theta$.

Weird twist thanks to quantum mechanics:

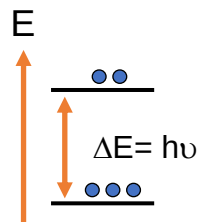
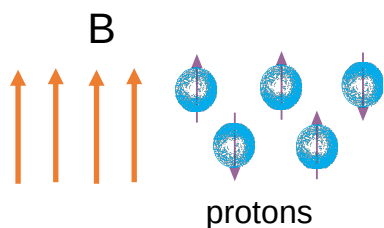
Only two observable energy states: "up" and "down"

But we only look at ensembles
Total E: add... practically a continuous variable...



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Magnetic moment in an external B field



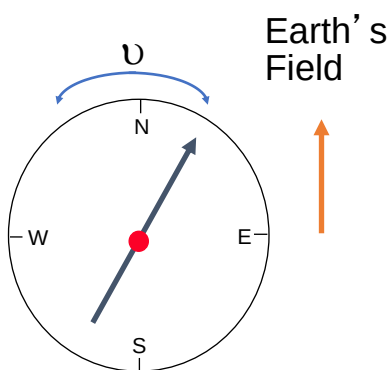
Thermal energy jostles some into high - energy state.

The net-alignment grows as ΔE gets bigger (higher field, stronger M) or T gets cold.

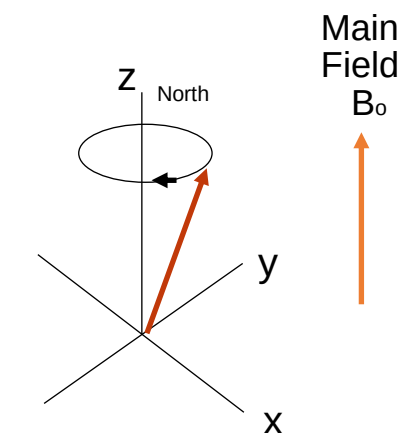
$$(N_{\uparrow} - N_{\downarrow})/N_{\text{TOT}} = 1 - \exp(-\Delta E/kT) \approx 10^{-4}$$

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Compass needles in B field: motion



$$\text{Freq} = \gamma B$$

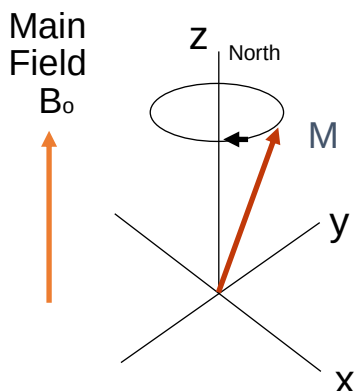


$$42.58 \text{ MHz/T}$$

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Gyroscopic motion:

Angular momentum
A downward (or upward) force)



- Proton has spin (angular momentum)



- Proton has magnetic moment that wants to align w/ North

>>gyroscopic precession

$$\omega = \gamma B_0$$

Larmor precession freq. = 42.58 MHz/T

The only equation you need to know...

M_z is a constant of motion (doesn't change)

M_{xy} goes round in simple circles

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Excitation:

Displacing the spins from equilibrium (North)

Problem:

M must be moving for us to detect it. Must initiate motion (oscillation)
must bring it away from equilibrium state...

Solution:

- 1) Tilt the magnet or compass suddenly.
- 2) Drive the magnetization (compass needle) with a periodic field.

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Excitation: Resonance

Why does only one frequency efficiently tip protons?

Resonant driving force.

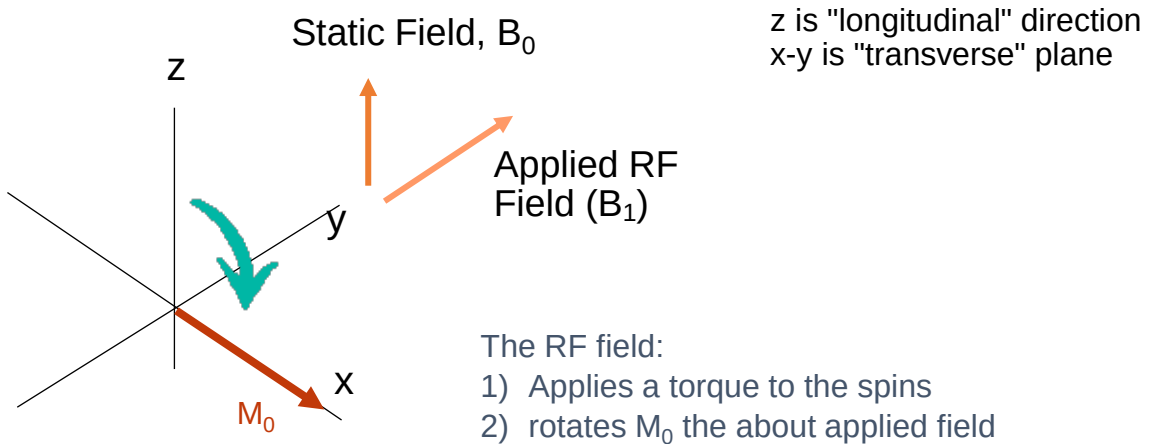
It's like pushing a child on a swing in time with the natural oscillating frequency.

Any other frequency would add half the time and subtract half the time...

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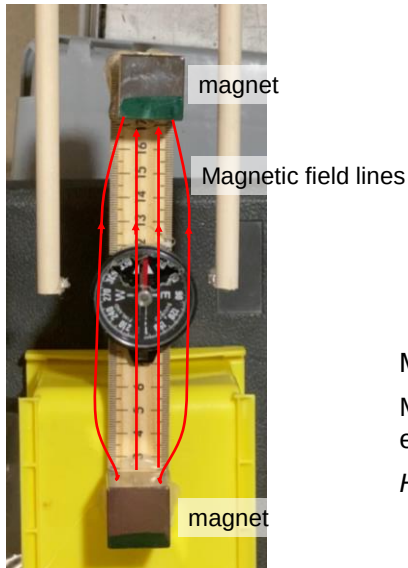
Excitation: Resonance

Why does only one frequency efficiently tip protons?



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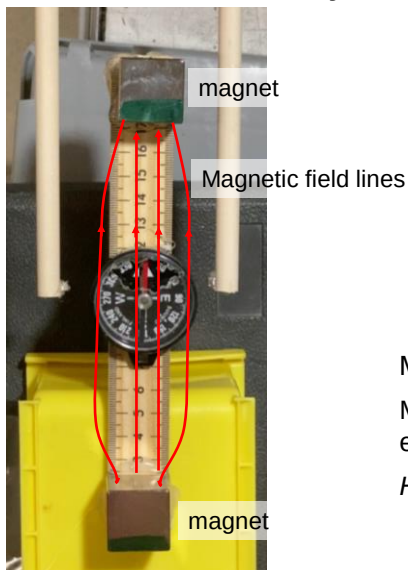
A little appreciated fact:
they wobble around north



MRI "sees" the wobble:
Moving magnet generates
electricity (generator effect)
How to get it started?

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A little appreciated fact:
they wobble around north

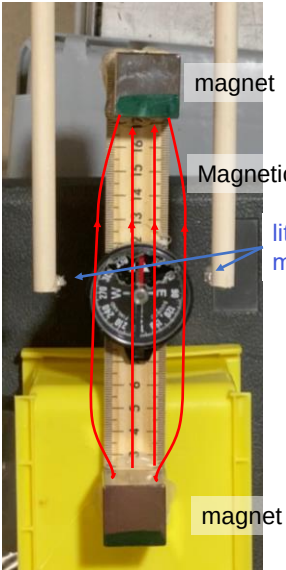


MRI "sees" the wobble:
Moving magnet generates
electricity (generator effect)
How to get it started?



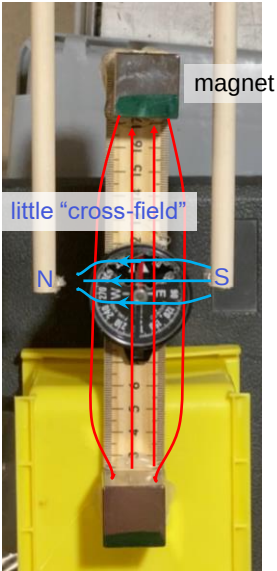
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Getting the wobble started:
“excitation”



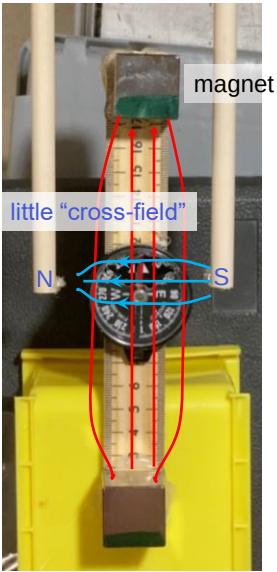
little magnets;
make a “cross-field”

This cross field is weak and just barely effects the compass.



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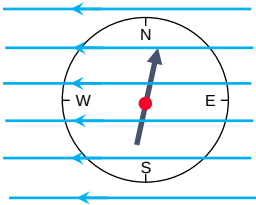
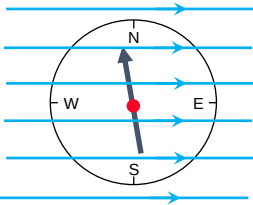
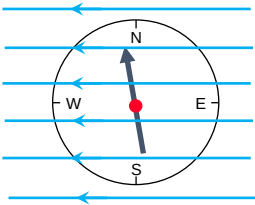
Getting the wobble started:
“excitation”



Once the little field has maximally perturbed needle, it starts to wobble back... *reverse the field*

This drives it back...

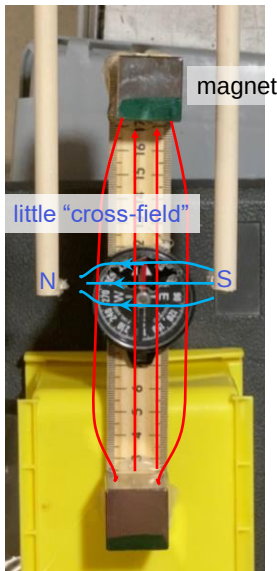
Repeat...



In MRI, the “excitation field” repeats this about 100,000 times!

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Getting the wobble started: “excitation”



Movie: spinning the “little magnet: with an electric drill...

When the drill speed matches the natural wobble frequency, it goes crazy...

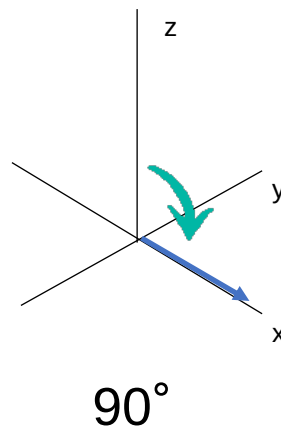
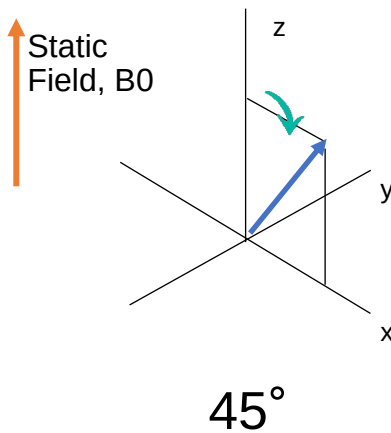
The “resonance” phenomena in Magnetic Resonance Imaging...



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Excitation: “tip angle”

Can tip it a little bit, or fully into xy plane...



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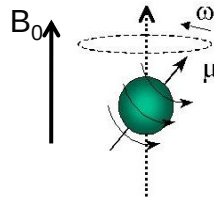
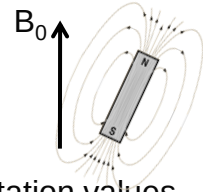
Aside on quantum mechanics

A single spin is a completely QM beast...

An ensemble of 10^{10} spins is a completely classical thing...

Voxel size: $0.1\text{mm} \times 0.1\text{mm} \times 0.1\text{mm}$ has 6×10^{16} protons
 >> classically described

- 1) Field produces torque on Magnetic moment
- 2) Angular momentum results in precession.
- 3) Equations of QM are the same, but for the average expectation values



$$\nu = \gamma B_0$$

Only equation needed...

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Detecting the precessing magnetization

Faraday's Law

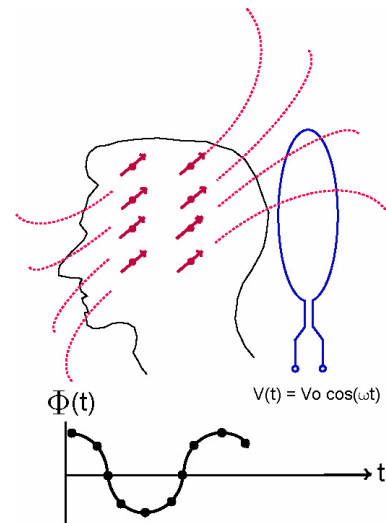
Generator effect:

A moving bar magnet induces a voltage in a coil of wire.

Flux: $\Phi = n B_{\text{spins}} A$

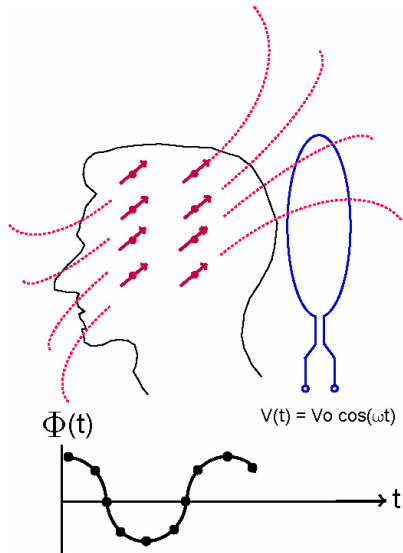
Induced voltage: $V(t) = -d\Phi/dt$

The RF coil design is the #1 determinant of the system SNR



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Detecting the precessing magnetization



Faraday's Law

$$\Phi(t) \propto M_0 \cos(\omega t)$$

$$V(t) = -\frac{d\Phi(t)}{dt}$$

$$V(t) \propto -M_0 \omega \sin(\omega t)$$

$$M_0 \propto B_0 \quad \omega \propto B_0$$

$$V(t) \propto B_0^2$$

$$\text{freq} = \gamma B_0 = 125\text{MHz at } 3\text{T}$$

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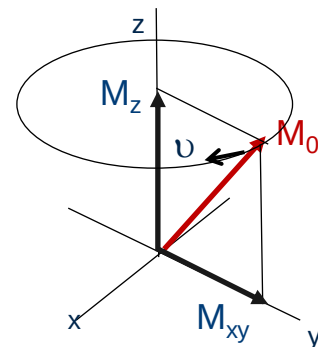
Two components of the Magnetization

One that moves, one that doesn't

M_z : Stationary, does not contribute to signal.
Tells you how much is "stored" ready for the next excitation.

M_{xy} : Processes at $\nu = \gamma B_{\text{total}}$.
Determines amplitude and phase of detected signal.

Image encoding: keep track of M_{xy} when a gradient is on...

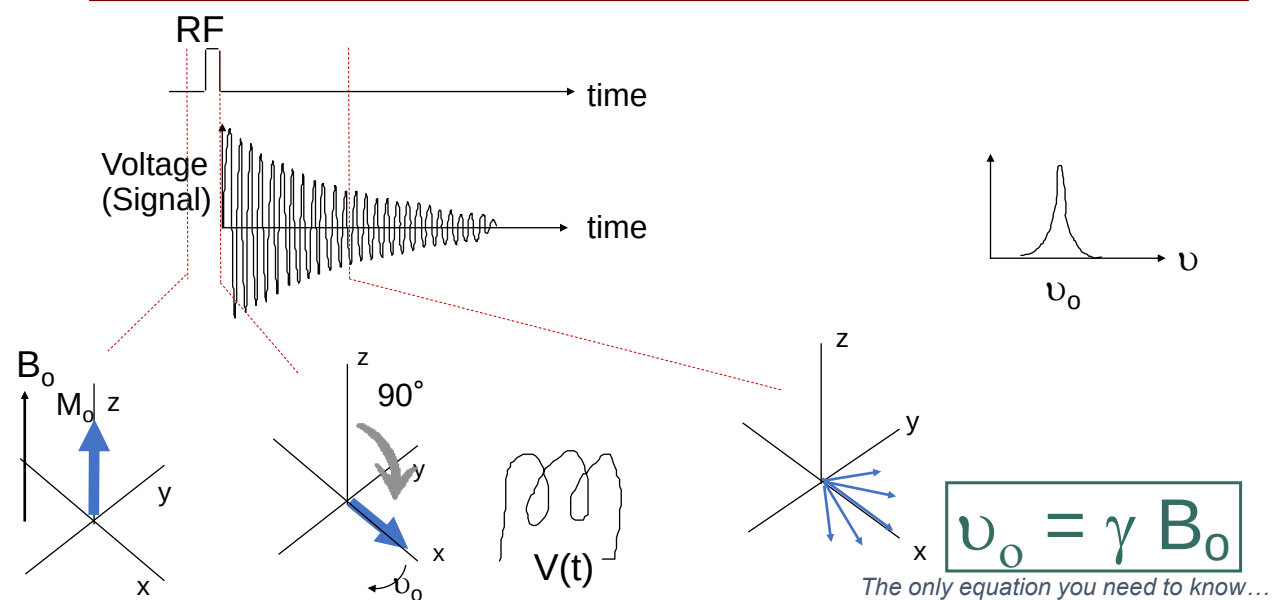


$$\nu_0 = \gamma B_0$$

The only equation you need to know...

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The NMR Signal: $M_{xy}(t)$

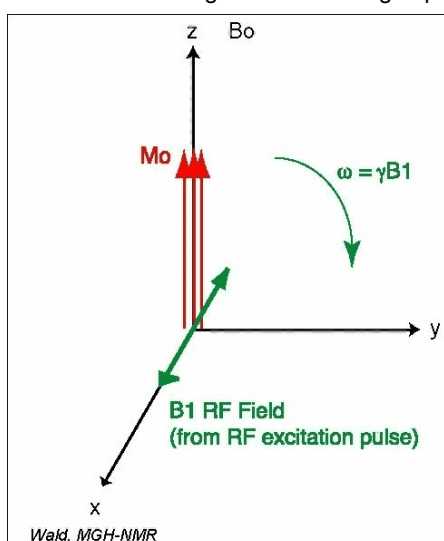


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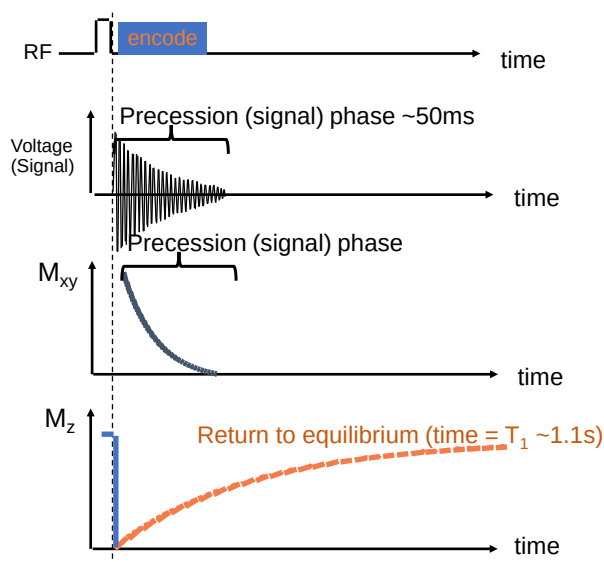
Three steps in MRI

And two major time-scales...

Animation in rotating frame of average spin...



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Physical Foundations of MRI

NMR: 60 year old phenomena that generates the signal from water.

MRI: using NMR signal to generate an image; map the water signal...

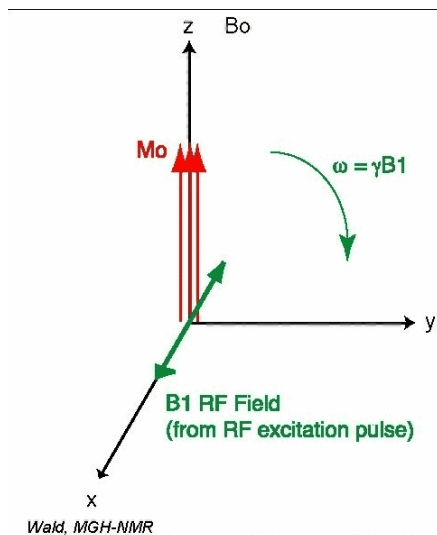
Three magnetic fields (generated by 3 coils)

- 1) static magnetic field B_0
- 2) RF field that excites the spins B_1
- 3) gradient fields that encode spatial info
 G_x, G_y, G_z

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Three steps in MRI

And two major time-scales...



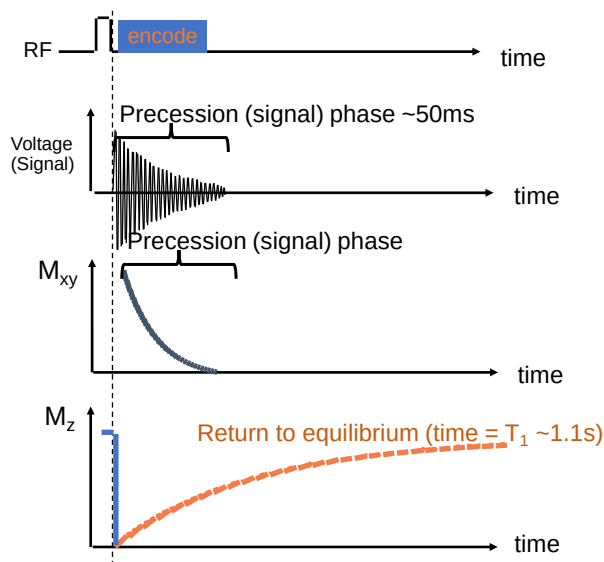
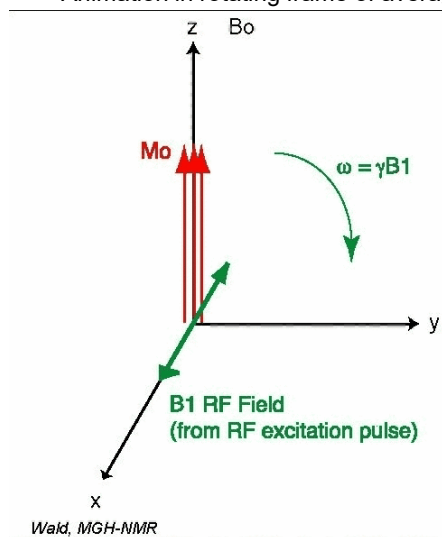
- 0) Equilibrium (magnetization points along B_0)
- 1) RF Excitation (tip magn. away from equil.)
- 2) Precession induces signal,
dephasing (timescale = T_2, T_2^*).
- 3) Return to equilibrium (timescale = T_1).

30

Three steps in MRI

And two major time-scales...

Animation in rotating frame of average spin...



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Three steps in MRI

And two major time-scales...

0) Equilibrium (magnetization points along B_0)

1) RF Excitation (tip magn. away from equil.)

2) Precession induces signal,
dephasing (timescale = T_2 , T_2^*).

3) Return to equilibrium (timescale = T_1).

proton
density
weighting

T_2 or T_2^*
weighting

T_1 Weighting

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Contrast in MRI: proton density

Encode image immediately after excitation

Tissue with more protons per cc gives more signal (brighter).

No chance to dephase thus:

- maximum signal strength
- No differences due to different tissue dephasing times (T2 values).

Magnetization starts fully relaxed (full M_z), thus no T1 differences either...

The most boring contrast...



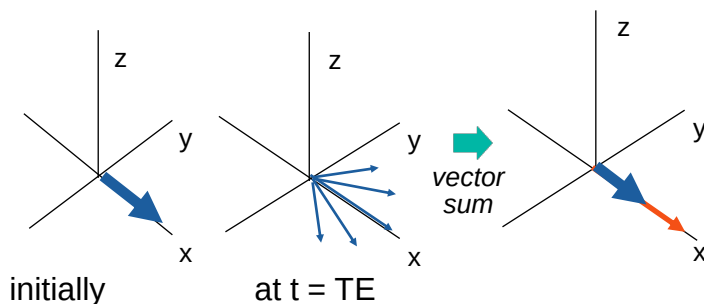
33

Contrast in MRI: T2* dephasing

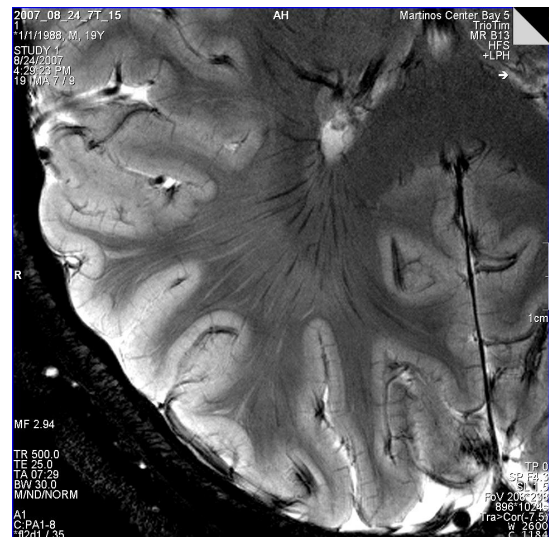
Wait a bit to encode image after excitation

Wait time TE before measuring M .

Shorter T2* spins have dephased



The iron and myelin rich tissues dephase faster (are dark)...

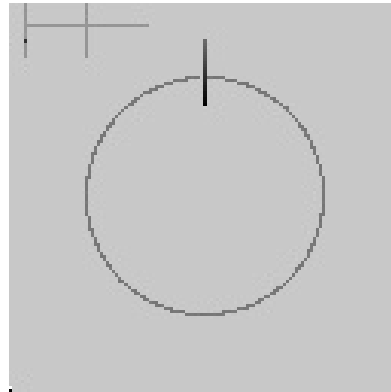


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Contrast in MRI: T_2^* dephasing

Ants on the race-track version

Just the tips of the
vectors...

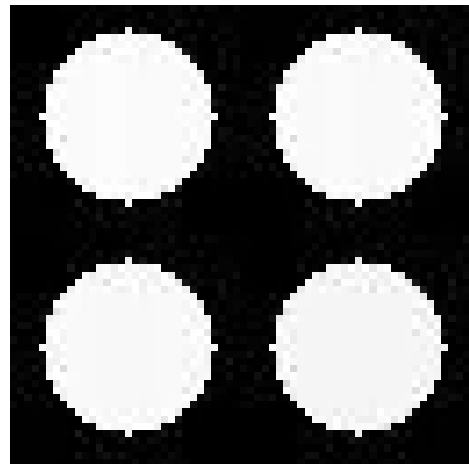


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Contrast in MRI: T_2^* dephasing

4 tubes, each with different T_2^*

Signal as a function
of time...



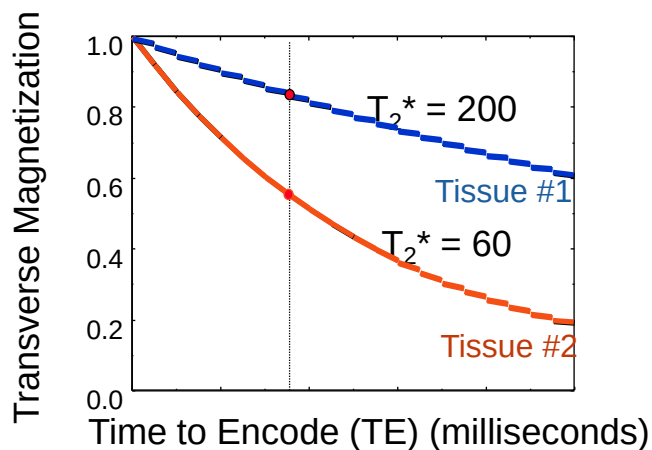
36

Contrast in MRI: T_2^* dephasing

The “contrast” is the signal difference

No contrast difference immediately after excitation -> must wait (but not too long!).

Choose TE for max. inten. difference.

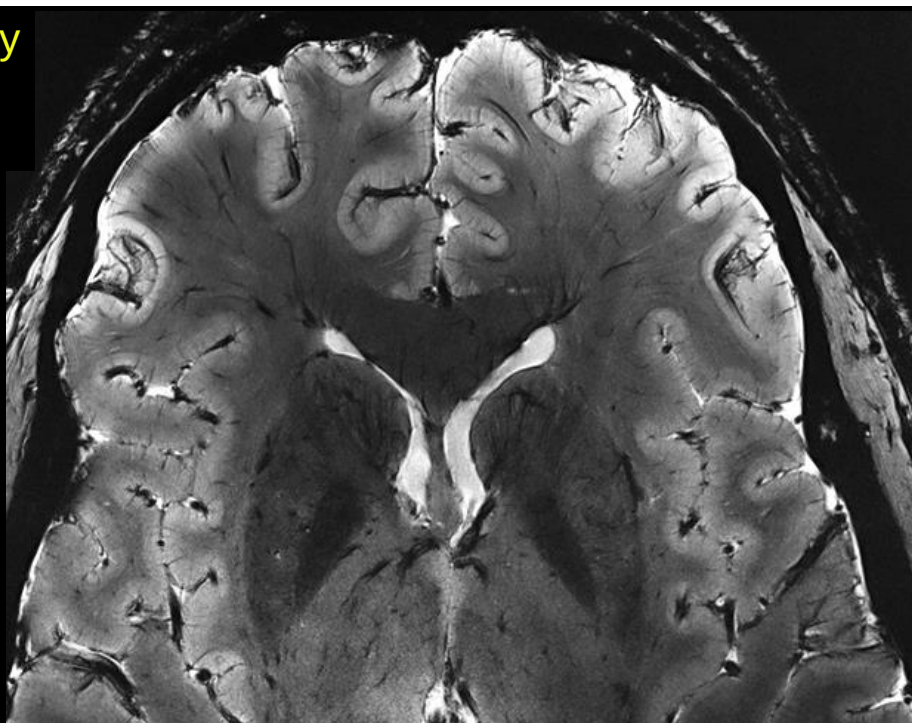


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7T 32ch MGH array

TE = 30ms

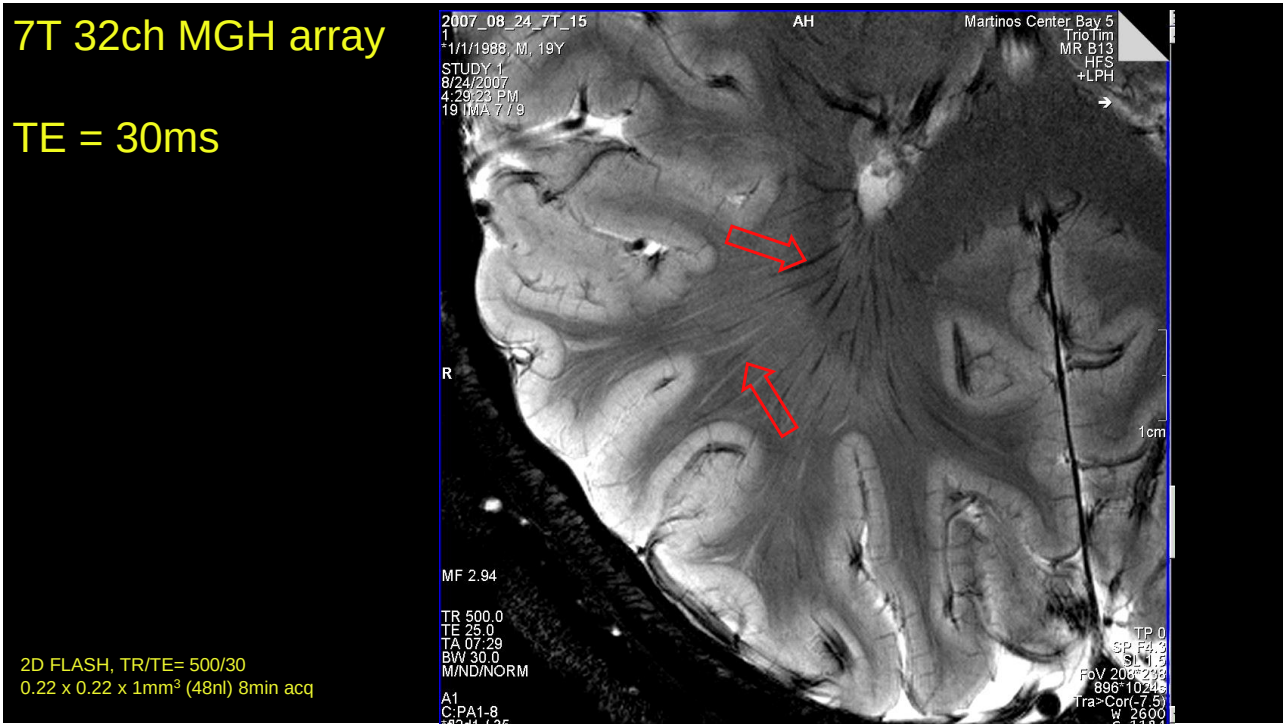
2D FLASH, TR/TE= 500/30
0.22 x 0.22 x 1mm³ (48nl) 8min acq



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7T 32ch MGH array

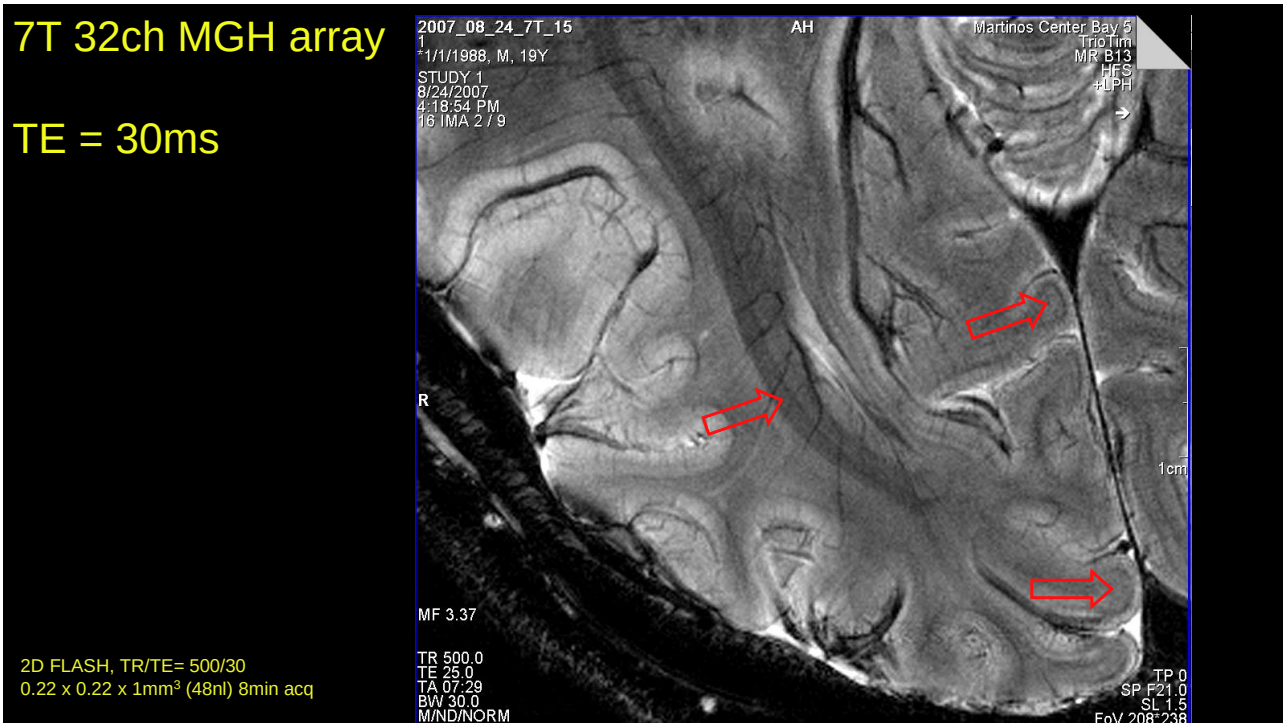
TE = 30ms



39

7T 32ch MGH array

TE = 30ms



40

7T 32ch

TE = 30ms

2D FLASH, TR/TE= 500/30
0.22 x 0.22 x 1mm³ (48nl) 8m



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Signal to Noise Ratio (SNR) in MRI

Need signal, not noise!

- Most important piece of hardware is the RF coil, next a big magnet...
- $\text{SNR} \propto \text{voxel volume (\# of spins)}$
- $\text{SNR} \propto \text{SQRT}(\text{total time of data collection})$
- SNR depends on the amount of signal you throw away to better visualize the brain (gain image contrast)

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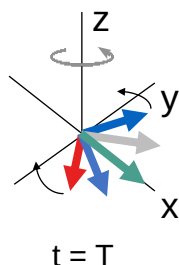
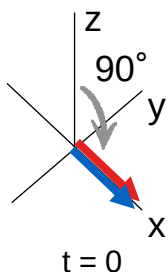
Can accelerate dephasing by applying a gradient

Dephasing is entirely from a spatial difference in the applied static fields under our control.



5 spins at different x locations: X

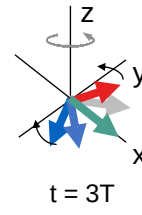
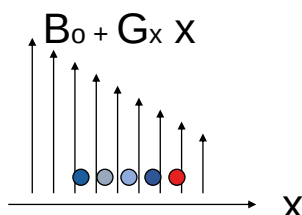
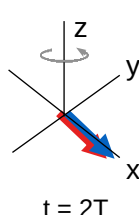
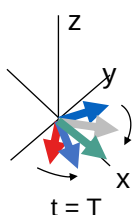
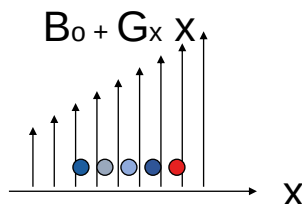
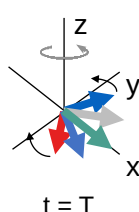
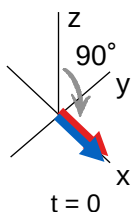
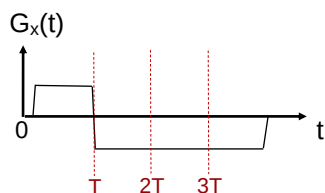
Red spin processes **fastest** due to its higher local field



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Parlor trick: gradient echo

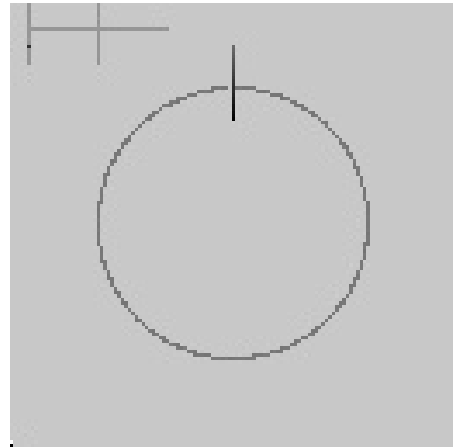
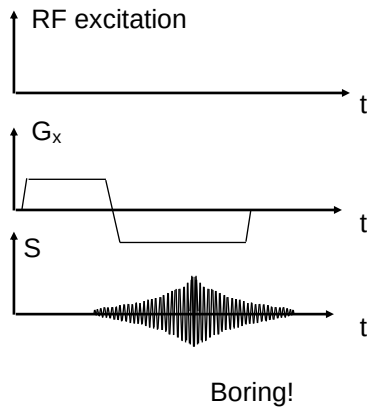
Amaze your friends; de-phase them then re-phase them!



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Parlor trick: gradient echo

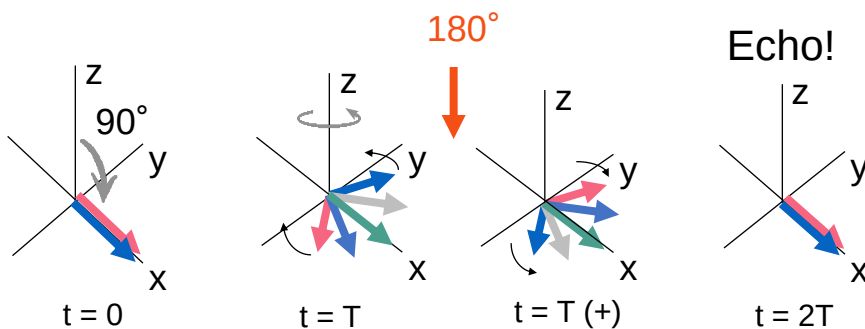
Ants on the racetrack version



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Parlor trick: spin echo

Re-phase even if dephasing is natural...

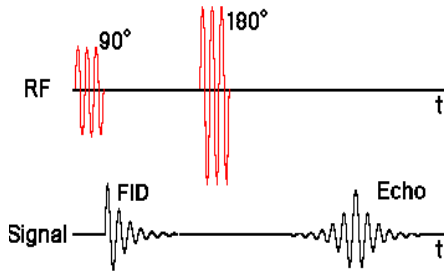


Blue arrows precesses faster due to local field inhomogeneity than red arrow

46

Spin Echo

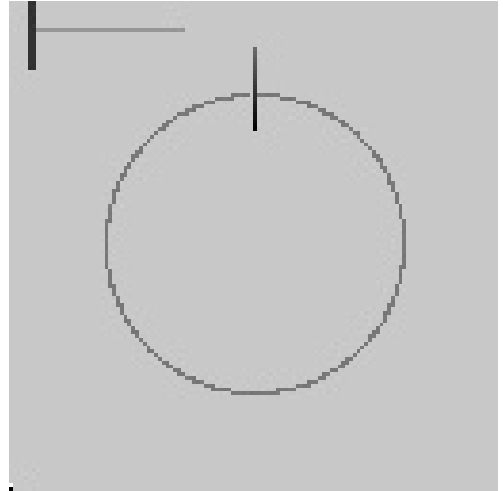
Re-phase even if dephasing is natural...



180° pulse only helps cancel **static** inhomogeneity

The “runners” can have static speed distribution.

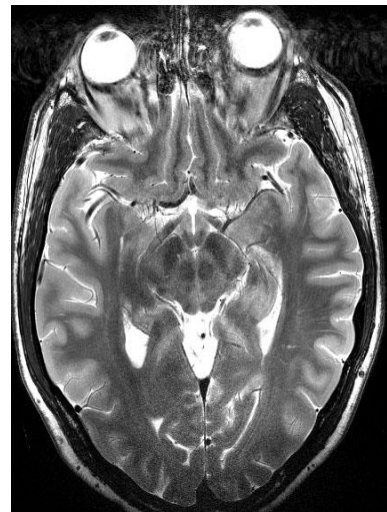
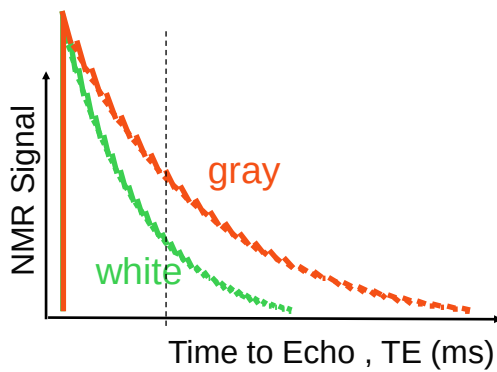
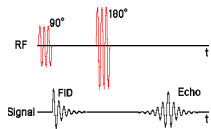
If a runner trips, will not make it back in phase with the others.



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Spin Echo

The most useful clinical contrast!



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T_1 contrast

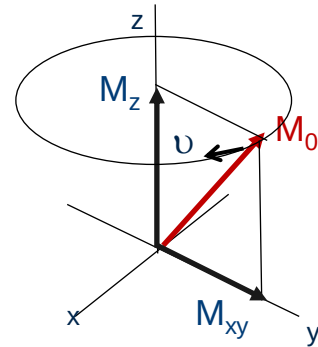
T_1 times (return to z axis)
differ across brain tissue by 10x...

M_z : Stationary, does not contribute to signal.

Tells you how much is "stored" ready for the next excitation.

M_{xy} : Processes at $\omega = \gamma B_{\text{total}}$.

Determines amplitude and phase of detected signal.

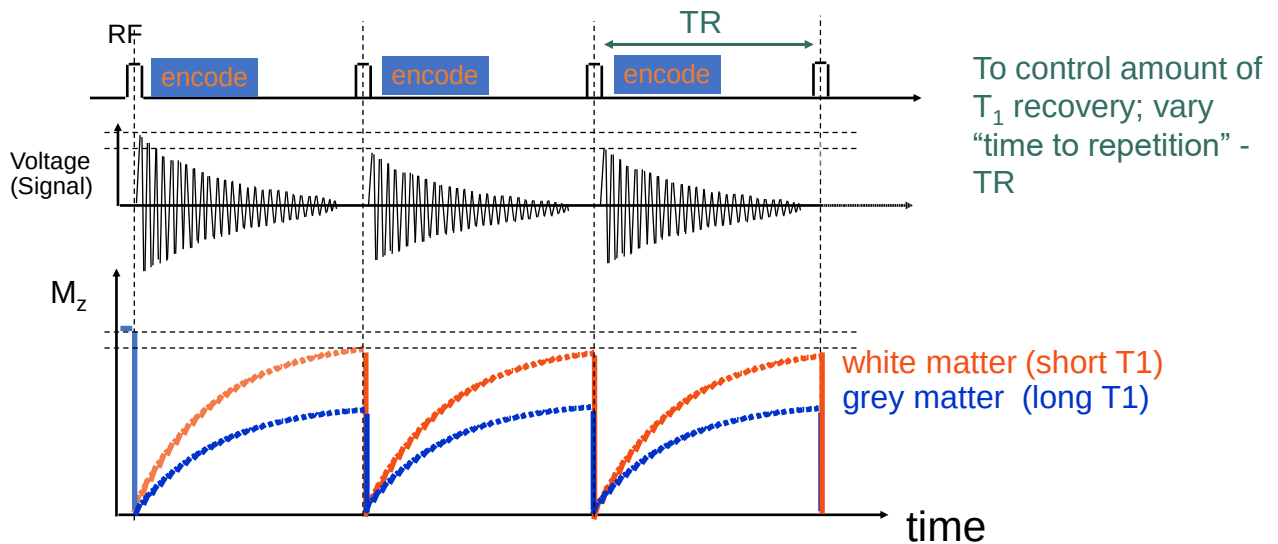


To control amount of T_1 recovery; vary "time to repetition" - TR

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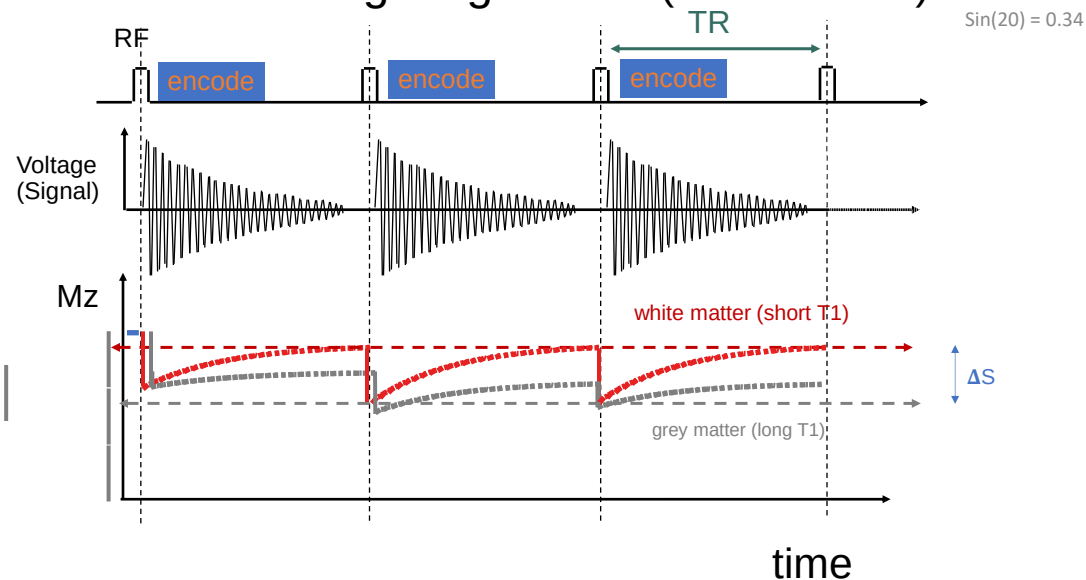
T_1 contrast

T_1 weighting w/ 90° excitation (saturation recovery)



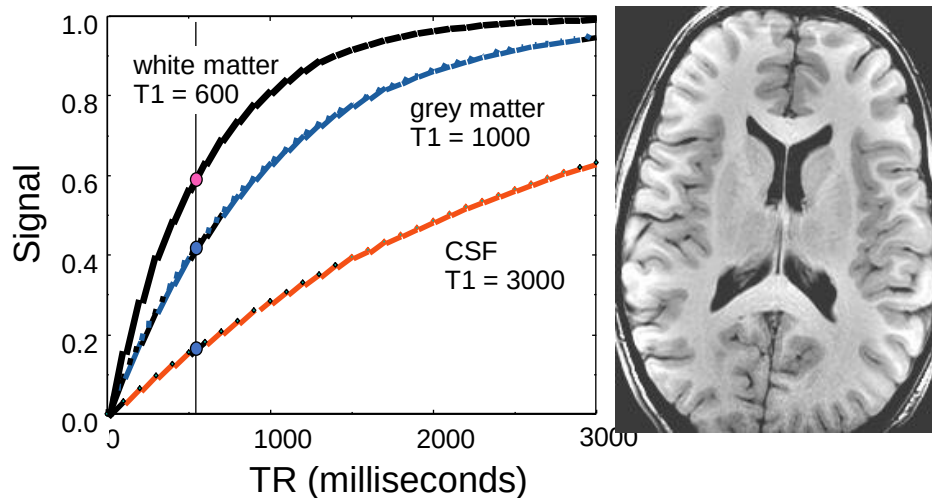
50

T1 weighting in MRI (w/ 20° excite)



51

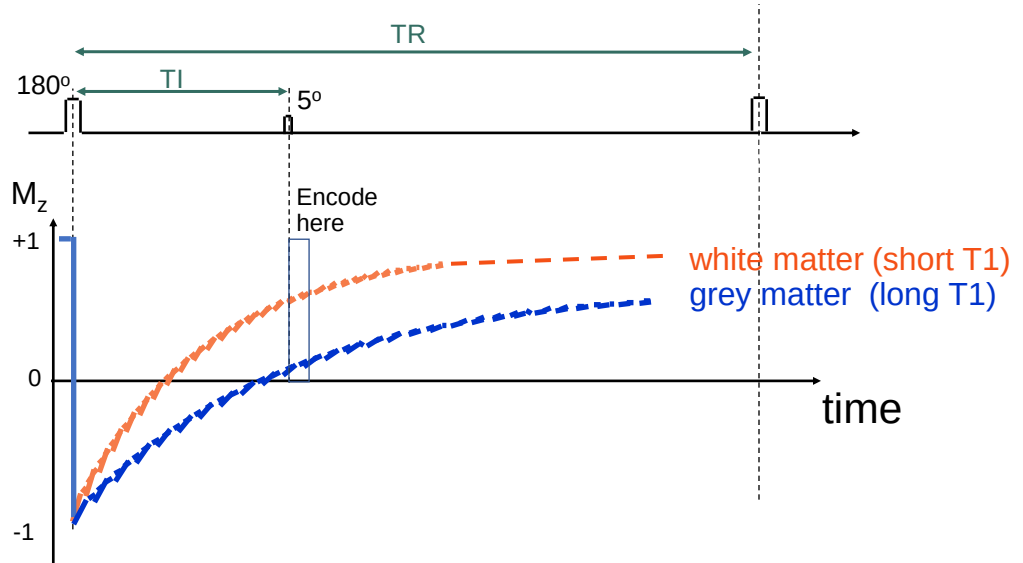
T1-Weighting



Wald, fMRI MR Physics

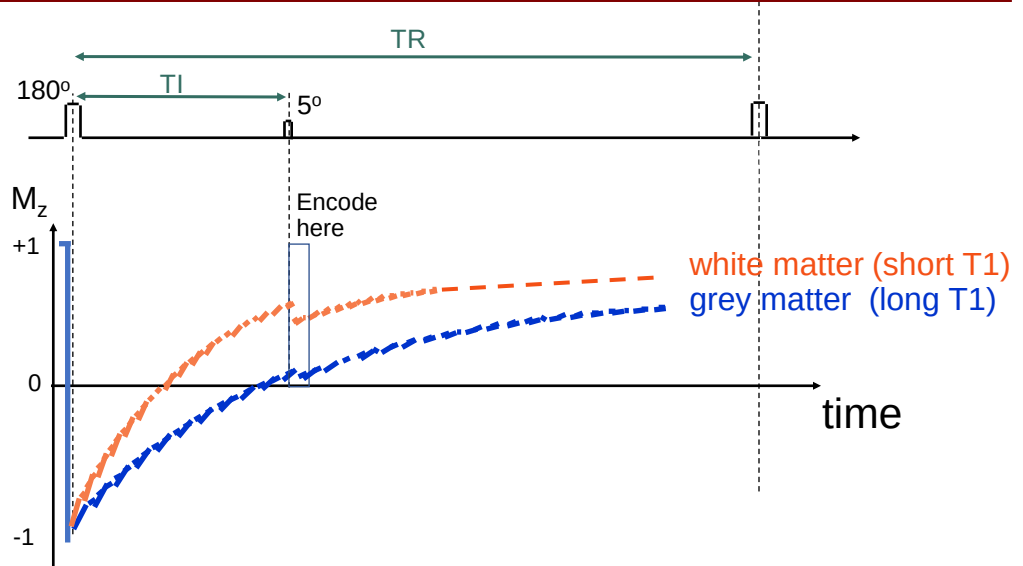
52

T1 weighting in MRI, Inversion recovery



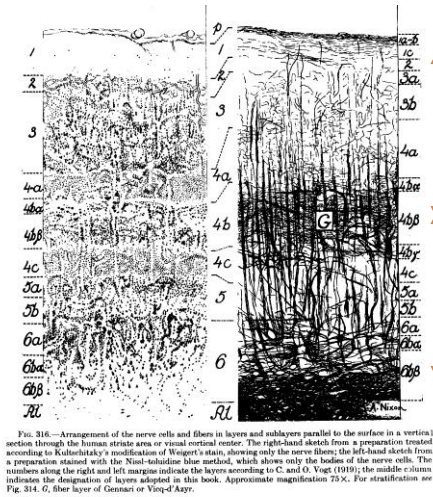
53

T1 weighting in MRI, Inversion recovery



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Source of T1 and T2 contrast in brain: Myelin content

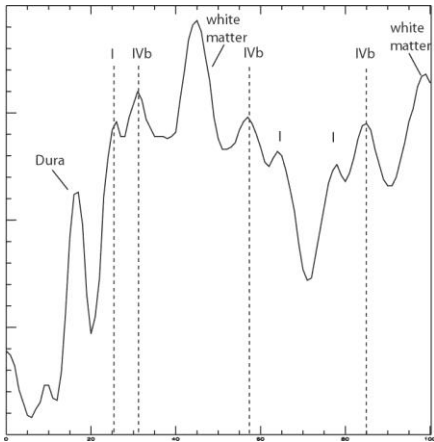
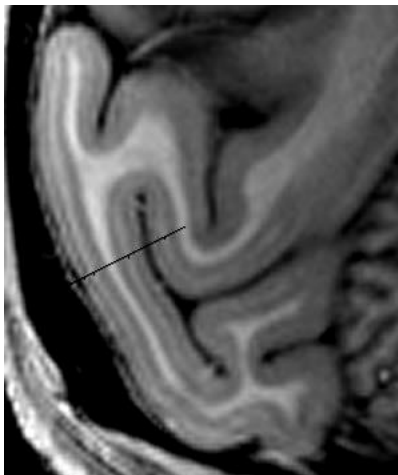


Nissel stain: cell bodies
Weigert stain: fibers

Determine functional boundaries based on MR strucure alone...

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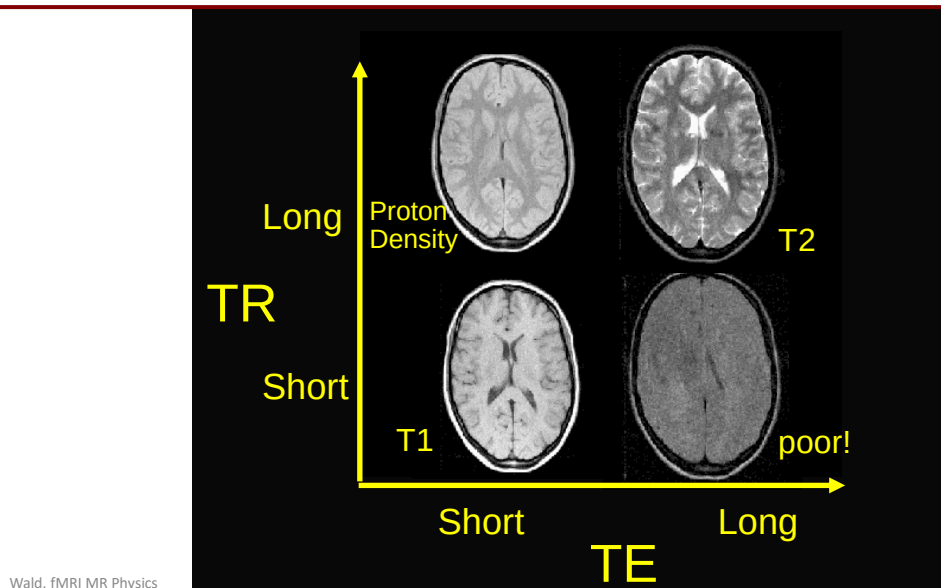
Source of T1 and T2 contrast in brain: Myelin content



Wald, fMRI MR Phys

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Image contrast summary: TR, TE



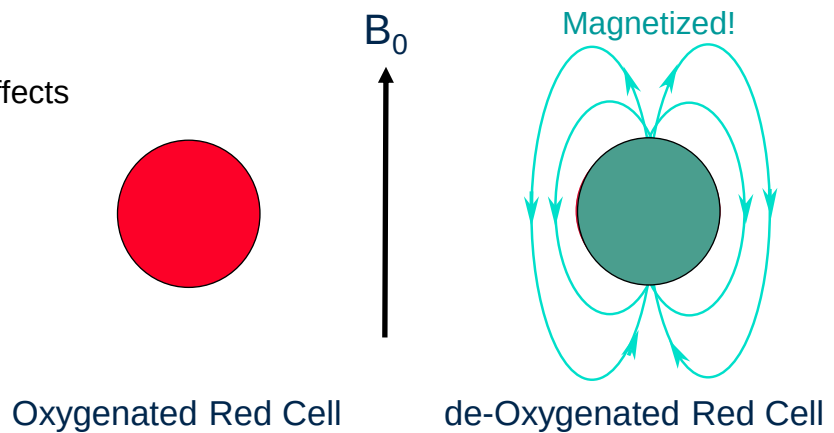
57

Basis of fMRI: BOLD contrast

Image signal change between activated and resting tissue

Hemodynamic Changes

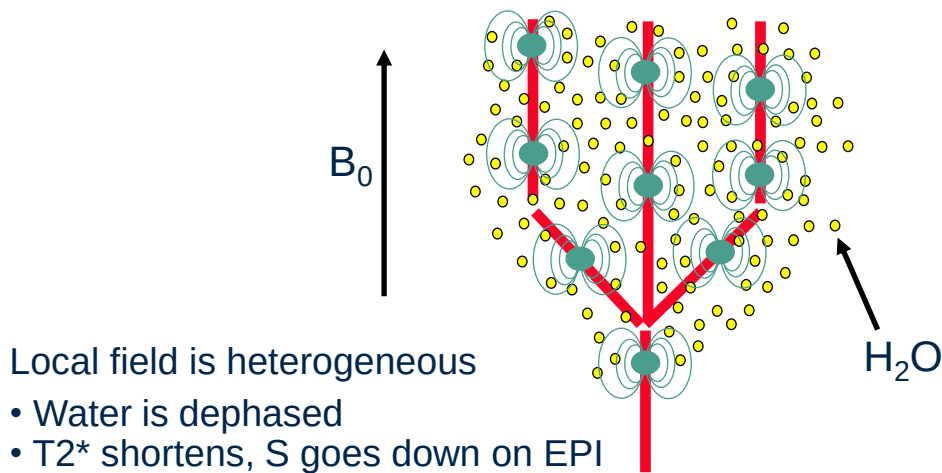
- Direct Flow effects
- Blood oxygenation effects



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Adding magnetic stuff to blood; T2* effects

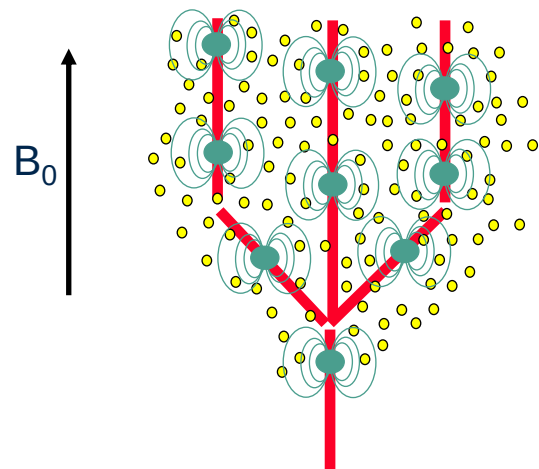
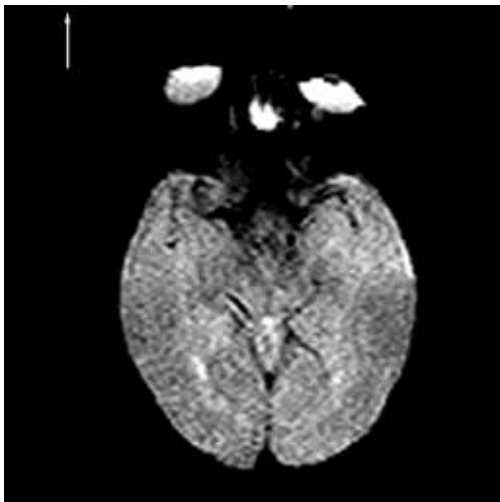
Add a paramagnetic agent (Gadolinium)



59

Adding magnetic stuff to blood; T2* effects

Add a paramagnetic agent (Gadolinium)



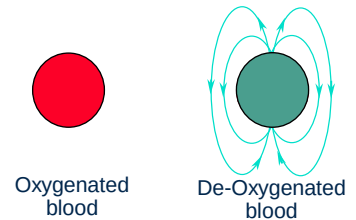
60

Physicist's analysis of BOLD

During activation:

- Tissue sucks more O₂ out of blood
- Blood in activated voxel is more de-oxygenated
- De-oxygenated = magnetic
- Magnetic stuff dephases water signal
- **Signal goes down during activation...**

Wrong!

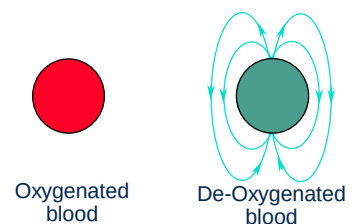


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Biologist's analysis of BOLD

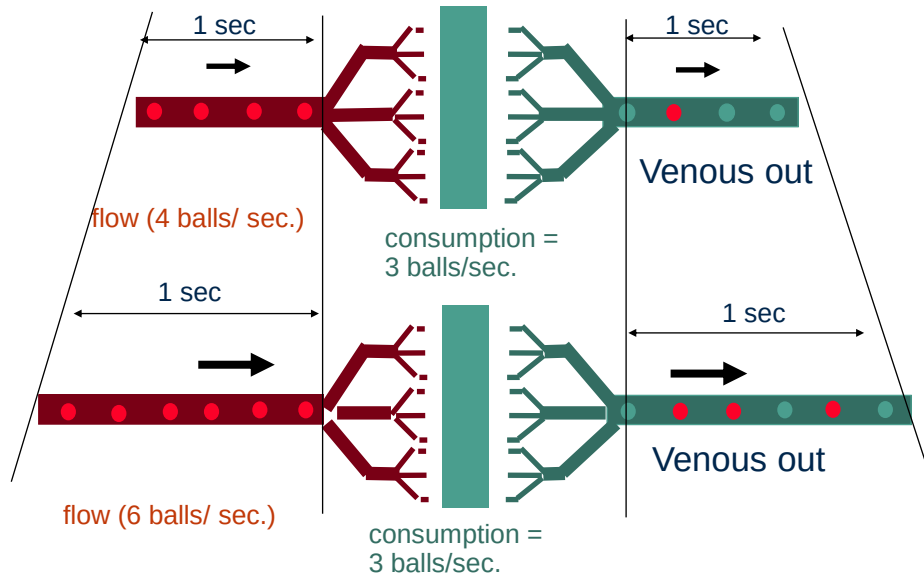
During activation:

- Tissue sucks only a little more O₂ out of blood
- Blood flow & volume goes up to meet need
- Oxygenated blood floods voxel
- Less magnetic stuff in blood to dephase water
- Signal goes **up** during activation...



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De-oxy Hb conc. goes down when flow goes up



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Blood oxygenation dynamics summary

During activation:

- Increases blood flow ($F \uparrow$)
- Increases blood volume ($V \uparrow$)
- Small increase in oxygen consumption

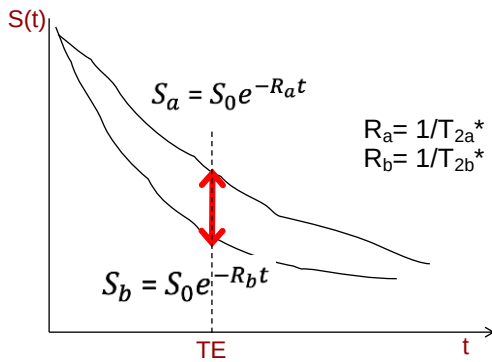
So:

venous $O_2 \uparrow$
 deoxy Hb concentration $\downarrow$
 less magnetic stuff
 less dephasing
 MR signal increases on T_2^* weighted image

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Optimize CNR as a function of TE

$$\text{SNR} = S/\sigma_n ; \quad \text{CNR} = \Delta S/\sigma_n$$



$$\Delta S = S_0 e^{-R_a t} - S_0 e^{-R_b t}$$

$$\frac{\Delta S}{S_0} = e^{-R_a t} - e^{-R_a t} e^{-\Delta R t}$$

$$\frac{\Delta S}{S_0} = e^{-R_a t} (1 - e^{-\Delta R t})$$

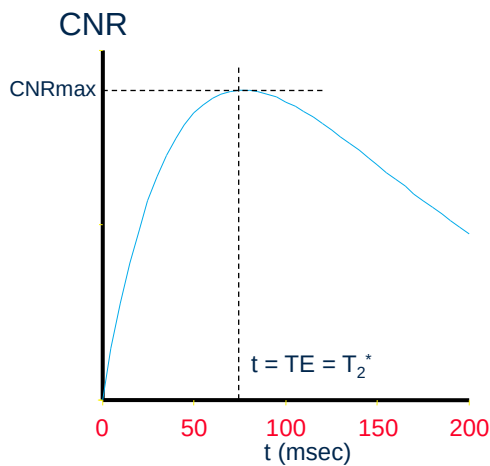
$$\frac{\Delta S}{S_0} = -\Delta R t e^{-R_a t}$$

$\Delta R \triangleq R_b - R_a$ $R_b = \Delta R + R_a$ $e^{-x} \approx 1 - x$

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CNR as a function of TE

$$\text{SNR} = S/\sigma_n ; \quad \text{CNR} = \Delta S/\sigma_n$$



$$\frac{\Delta S}{S_0} = -\Delta R t e^{-R_a t}$$

$$\text{Find max: } \frac{\partial}{\partial t} \left(\frac{\Delta S}{S_0} \right) = 0$$

$$t_{opt} = 1/R_a = T_2^*$$

$$\text{So use } TE = t_{opt} = T_2^*$$

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Other MR methods for mapping activation

98% of all human studies done with BOLD
Many animal studies done with Fe injected CBV

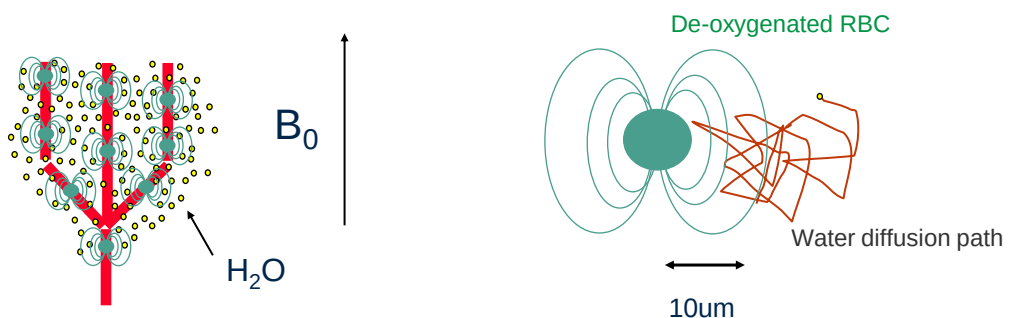
Any one of the following can be detected with MR:

Flow (CBF) $\uparrow\uparrow$	Increases signal on arterial spin labeled (ASL) flow scans
DeoxyHb $\downarrow$	Increases signal on " T_2/T_2^* -weighted" scans ("BOLD" method)
Blood Vol. (CBV) $\uparrow\uparrow$	Decreases signal w/ contrast agent is on board (IRON method).

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Paramagnetic agent in blood: T_2 change in BOLD

If the local fields were static, a spin echo would refocus them



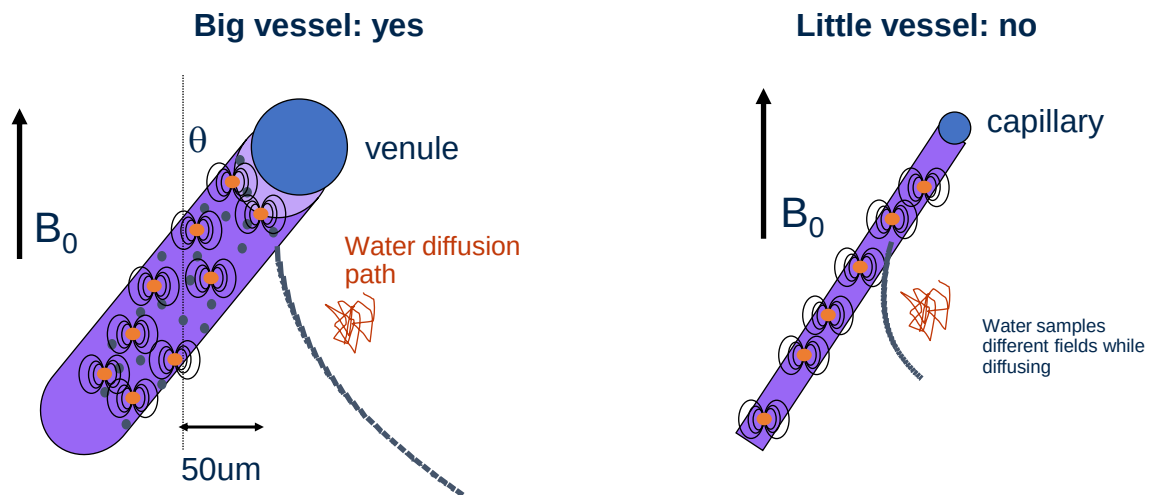
Diffusion through local fields: dynamic phase changes not refocused by spin echo

- T_2 shortens, S goes down on spin echo EPI
- T_2 effect increases with B_0^2

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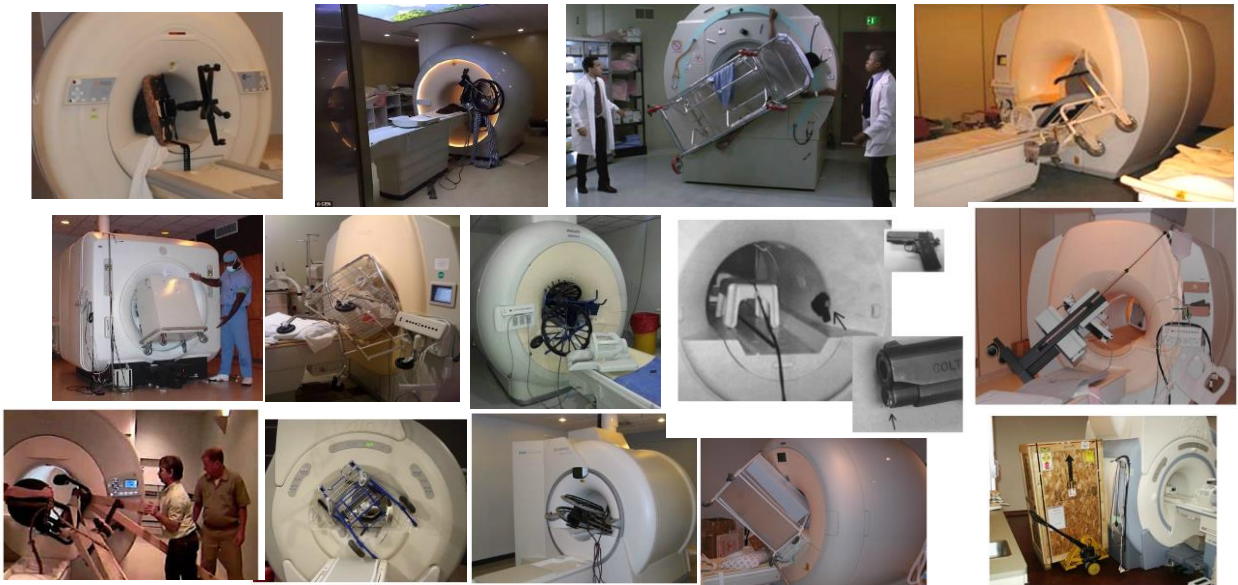
Paramagnetic agent in blood: whole vessel magnetizes

Is the field outside "magnetized" venule ~constant on length scale of water diffusion (~25um)?



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Thanks for your attention & vigilance!



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